Granular Instrumental Variables

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Aggregate shocks confound firm effects

The Goal: Estimate causal elasticity of wages to labor demand

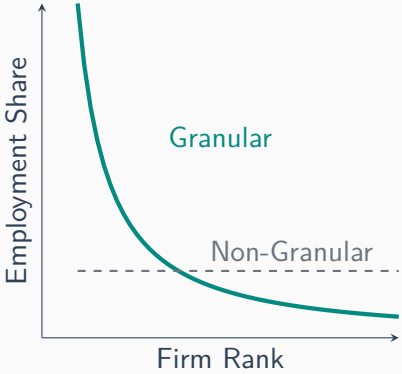
The Identification Challenge: **Simultaneity**

- **Supply:** Wages respond to labor demand and macro supply shocks
- **Demand:** Labor demand responds to wages and macro demand shocks

LLN fails in granular economies

1. In a standard economy, micro-shocks cancel out at a rate of $1/\sqrt{N}$ ▸ Proof

2. In a **granular** economy, size distribution of firms follows a power law (i.e., Zipf's Law: size $\propto \frac{1}{\text{rank}}$) $\implies$ large firms' idiosyncratic shocks drive aggregate outcomes



A Simple Model

Simultaneity prevents OLS estimation

Economy with N firms. System is given by:

$$w_t = \psi L_{St} + \varepsilon_t \tag{Supply}$$

$$L_{it} = \phi^d w_t + \eta_t + u_{it} \tag{Demand}$$

Where:

- w_t - wage (observed)
- L_{it} - number of workers at firm i (observed)
- η_t and ε_t - aggregate shocks (unobserved)
- u_{it} - idiosyncratic shock to firm i (unobserved)
- Main assumption: $corr(u_{it}, \eta_t) = 0,\ corr(u_{it}, \varepsilon_t) = 0$ ▸ Assumptions

We cannot estimate ψ (labor supply elasticity) and ϕ^d (labor demand elasticity) by OLS, due to simultaneity.

GIV differences size and equal weights

GIV exploits difference between size-weighted and equal-weighted averages.

- Define equal-weighted average:

$$L_{Et} = \frac{1}{N} \sum_{i=1}^{N} L_{it}$$

- Define size-weighted average:

$$L_{St} = \sum_{i=1}^{N} S_i L_{it}, \quad \sum_{i=1}^{N} S_i = 1$$

GIV is the difference between them:

$$z_t := L_{\Gamma t} = L_{St} - L_{Et} \tag{GIV}$$

GIV isolates idiosyncratic shocks

Substitute demand equation into averages: Proof

$$L_{St} = \phi^d w_t + \eta_t + u_{St}$$
$$L_{Et} = \phi^d w_t + \eta_t + u_{Et}$$

Common components ($\phi^d w_t + \eta_t$) difference out:

$$z_t = u_{St} - u_{Et} = \sum_{i=1}^{N} S_i u_{it} - \frac{1}{N}\sum_{i=1}^{N} u_{it} = \sum_{i=1}^{N}(S_i - \frac{1}{N})u_{it} \qquad \text{(GIV)}$$

z_t is made *only* of idiosyncratic shocks.

GIV enables 2SLS estimation

IV Estimation:

- **First Stage:** Regress L_{St} on z_t. Extracts $\widehat{L}_{St}$, the labor demand variation driven *only* by idiosyncratic shocks.
- **Second Stage:** Regress w_t on $\widehat{L}_{St}$ to obtain unbiased ψ.

Fundamental IV assumptions:

- **Relevance** (Cov$(L_{St}, z_t) \neq 0$):
  Idiosyncratic granular shocks must move aggregate market. This requires a concentrated market (e.g., high Herfindahl index).
- **Exclusion Restriction** (Cov$(z_t, \varepsilon_t) = 0$, Cov$(z_t, \eta_t) = 0$):
  Idiosyncratic granular shocks must not directly affect aggregate wages through any channel, i.e. they are orthogonal to macro shocks ($z_t \perp \eta_t, \varepsilon_t$).

GIV yields both elasticities

By exogeneity, $\mathbb{E}[u_t \varepsilon_t] = 0 \implies \mathbb{E}[z_t \varepsilon_t] = 0$, where $u_t = (u_{it})_{i=1,\ldots,N}$.

Using supply equation:

$$w_t = \psi L_{St} + \varepsilon_t \implies \mathbb{E}[(w_t - \psi L_{St}) z_t] = 0 \implies \psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]}$$

Using equal-weighted demand equation: Proof

$$\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = 0 \implies \phi^d = \frac{\mathbb{E}[L_{Et} z_t]}{\mathbb{E}[w_t z_t]}$$

The Full Model

Handling heterogeneous factor loadings

Heterogeneous loadings bias simple GIV

We now allow for heterogeneous loadings λ_i:

$$L_{it} = \phi w_t + \lambda_i \eta_t + u_{it}$$

The simple GIV assumes every firm has the same loading $\lambda_i = 1$.

If large firms are more sensitive to the Fed, the simple GIV becomes: Proof

$$z_t = (\lambda_S - \lambda_E)\eta_t + (u_{St} - u_{Et})$$

If $\lambda_S \neq \lambda_E$, the instrument is **invalid** ($\mathbb{E}[z_t \eta_t] \neq 0$, i.e., the Exclusion Restriction fails).

Orthogonal weights restore exclusion

Substitute demand equation into definition of weighted GIV:

$$\begin{aligned} z_t &= \sum \Gamma_i L_{it} = \sum \Gamma_i (\phi w_t + \lambda_i \eta_t + u_{it}) \\ &= \phi w_t (\sum \Gamma_i) + \eta_t (\sum \Gamma_i \lambda_i) + \sum \Gamma_i u_{it} \end{aligned}$$

Where Γ is the vector of weights that we want to find.

To ensure z_t contains *only* idiosyncratic shocks (u_{it}), we must satisfy two conditions:

1. $\sum \Gamma_i = 0$
2. $\sum \Gamma_i \lambda_i = 0$

Projection finds optimal weights

For this, we use $N \times N$ projection matrix Q, orthogonal to factor loadings ($Q\lambda = 0$):

Refresher

$$Q = I - \lambda(\lambda'\lambda)^{-1}\lambda'$$

Optimal weights are then given by $\Gamma' = S'Q$.

- Intuitively, we want Γ to be as close as possible to S but satisfy $\Gamma \perp \lambda$
- This is a standard projection problem: $\Gamma = S - \lambda(\lambda'\lambda)^{-1}\lambda'S$

Propositions

Application

GIV in the Israeli Labor Market

Israeli tech is highly granular

Israeli high-tech industry is dominated by a few multinational R&D centers.

- Intel, Google, Microsoft, NVIDIA account for disproportionate share of employment
- Size distribution of tech in Israel exhibits strong power law (high Herfindahl index h)
- Difficult to separate Israeli macro shocks (wars, interest rates) from hiring dynamics

GIV unconfounds tech spillovers

How do we isolate causal spillover of tech giants on wages?

- Exploit global shocks specific to multinationals (u_{it}) that are exogenous to Israeli economy (global restructuring at Intel, worldwide AI hiring boom at NVIDIA).
- Construct GIV by taking size-weighted employment shocks of tech giants and projecting out common Israeli macroeconomic factor loadings.
- Regress aggregate Israeli wages on component of labor demand driven *only* by these isolated, granular tech shocks.
- Measure exactly how much a hiring freeze at a major tech firm depresses aggregate local wages, separated from general economic downturns.
- This is perfectly suited for small, open economies like Israel, where granular actors often drive aggregate volatility.

Appendix A

Derivations

Assumptions

The Central Identification Assumption

Individual shocks u_{it} are truly idiosyncratic—uncorrelated with aggregate shocks (η_t, ϵ_t) and controls (C_t^y, C_t^p):

$$u_t \perp \eta_t, \epsilon_t, C_t^y, C_t^p \tag{1}$$

Baseline Simplifications

A1 Factor loadings λ_i are known (relaxed in §IV.B).

A2 Controls have factor structure: $C_{it}^y = \lambda_i c_t^y$.

Error Structure

A3 Homoskedastic and uncorrelated across i: $\mathbb{E}[u_t u_t'] = \sigma_u^2 I$.

A4 General heteroskedasticity V^u (relaxed for optimality).

Note: All random variables are assumed to have finite fourth moments and are i.i.d. across periods.

Back

Equal weights decay at $1/\sqrt{N}$

Suppose there are N firms, each receiving an independent idiosyncratic shock u_{it}. Assume these shocks have mean 0 and a common variance σ^2:

$$\mathbb{E}[u_{it}] = 0, \quad \text{Var}(u_{it}) = \sigma^2, \quad \text{Cov}(u_{it}, u_{jt}) = 0 \text{ for } i \neq j$$

In a non-granular economy, the aggregate shock is an **equal-weighted** average:

$$u_{Et} = \frac{1}{N} \sum_{i=1}^{N} u_{it}$$

The variance of the aggregate shock is:

$$\text{Var}(u_{Et}) = \text{Var}\left(\frac{1}{N} \sum u_{it}\right) = \frac{1}{N^2} \sum \text{Var}(u_{it}) = \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N}$$

The aggregate volatility is therefore: $\sigma_{agg} = \frac{\sigma}{\sqrt{N}}$

Back

Size-weighting aggregates shocks

The size-weighted outcome is defined as $L_{St} = \sum_{i=1}^{N} S_i L_{it}$, where S_i are weights such that $\sum S_i = 1$.

$$L_{St} = \sum_{i=1}^{N} S_i(\phi^d w_t + \eta_t + u_{it})$$

$$L_{St} = \sum_{i=1}^{N} S_i \phi^d w_t + \sum_{i=1}^{N} S_i \eta_t + \sum_{i=1}^{N} S_i u_{it}$$

Since ϕ^d, w_t, and η_t do not depend on the index i, we can pull them out:

$$L_{St} = \phi^d w_t \left(\sum_{i=1}^{N} S_i \right) + \eta_t \left(\sum_{i=1}^{N} S_i \right) + \sum_{i=1}^{N} S_i u_{it}$$

Using the property $\sum S_i = 1$ and defining the size-weighted idiosyncratic shock as $u_{St} = \sum S_i u_{it}$, we get:

$$L_{St} = \phi^d w_t + \eta_t + u_{St}$$

Back

Equal-weighting averages shocks

The equal-weighted outcome is defined as $L_{Et} = \frac{1}{N}\sum_{i=1}^{N} L_{it}$.

$$L_{Et} = \frac{1}{N}\sum_{i=1}^{N}(\phi^d w_t + \eta_t + u_{it})$$

$$L_{Et} = \frac{1}{N}\sum_{i=1}^{N}\phi^d w_t + \frac{1}{N}\sum_{i=1}^{N}\eta_t + \frac{1}{N}\sum_{i=1}^{N}u_{it}$$

Again, pulling constants out of the summations:

$$L_{Et} = \phi^d w_t\left(\frac{1}{N}\sum_{i=1}^{N}1\right) + \eta_t\left(\frac{1}{N}\sum_{i=1}^{N}1\right) + \frac{1}{N}\sum_{i=1}^{N}u_{it}$$

Since $\frac{1}{N}\sum_{i=1}^{N}1 = \frac{1}{N}\cdot N = 1$, and defining the equal-weighted idiosyncratic shock as $u_{Et} = \frac{1}{N}\sum u_{it}$, we get:

$$L_{Et} = \phi^d w_t + \eta_t + u_{Et}$$

GIV identifies supply elasticity

To find ψ, we use z_t as an instrument for L_{St} in the supply equation. We take the expectation of the product of the supply equation and the instrument:

$$\mathbb{E}[w_t z_t] = \mathbb{E}[(\psi L_{St} + \epsilon_t) z_t]$$
$$\mathbb{E}[w_t z_t] = \psi \mathbb{E}[L_{St} z_t] + \mathbb{E}[\epsilon_t z_t]$$

By the exogeneity condition, $\mathbb{E}[\epsilon_t z_t] = 0$. Thus:

$$\psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]} \tag{2}$$

This is the standard IV formula where z_t instruments for L_{St}.

Back

GIV identifies demand elasticity

To find ϕ^d, we use the equal-weighted aggregate demand equation derived on slide 8:

$$L_{Et} = \phi^d w_t + \eta_t + u_{Et} \tag{41}$$

We aim to show that $\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = 0$. Substituting the demand equation:

$$\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = \mathbb{E}[(\eta_t + u_{Et}) z_t] = \mathbb{E}[\eta_t z_t] + \mathbb{E}[u_{Et} z_t] \tag{3}$$

We already know $\mathbb{E}[\eta_t z_t] = 0$. Now we examine $\mathbb{E}[u_{Et} z_t]$. Under the assumption that u_{it} are i.i.d. with variance σ_u^2:

$$\begin{aligned} \mathbb{E}[u_{Et} z_t] &= \mathbb{E}[u_{Et}(u_{St} - u_{Et})] \\ &= \mathbb{E}[u_{Et} u_{St}] - \mathbb{E}[u_{Et}^2] \end{aligned}$$

Back

GIV identifies demand elasticity

$$\mathbb{E}[u_{Et}u_{St}] - \mathbb{E}[u_{Et}^2] =$$

$$= \mathbb{E}\left[\left(\frac{1}{N}\sum_i u_{it}\right)\left(\sum_j S_j u_{jt}\right)\right] - \mathbb{E}\left[\left(\frac{1}{N}\sum_i u_{it}\right)^2\right]$$

$$= \frac{1}{N}\sum_i S_i\sigma_u^2 - \frac{1}{N^2}\sum_i \sigma_u^2$$

$$= \frac{1}{N}\sigma_u^2(1) - \frac{1}{N}\sigma_u^2 = 0$$

Since $\mathbb{E}[u_{Et}z_t] = 0$, it follows that $\mathbb{E}[(L_{Et} - \phi^d w_t)z_t] = 0$. Rearranging:

$$\mathbb{E}[L_{Et}z_t] = \phi^d\mathbb{E}[w_t z_t]$$

$$\phi^d = \frac{\mathbb{E}[L_{Et}z_t]}{\mathbb{E}[w_t z_t]}$$

Simple GIV leaks macro shocks

The simple GIV is defined as $z_t^{simple} = L_{St} - L_{Et}$. Let's derive its components:

1. **Size-weighted average (L_{St}):**

$$\begin{aligned} L_{St} &= \sum S_i L_{it} = \sum S_i(\phi w_t + \lambda_i \eta_t + u_{it}) \\ &= \phi w_t \underbrace{\sum S_i}_{1} + \eta_t \underbrace{\sum S_i \lambda_i}_{\lambda_S} + \underbrace{\sum S_i u_{it}}_{u_{St}} \\ &= \phi w_t + \lambda_S \eta_t + u_{St} \end{aligned}$$

2. **Equal-weighted average (L_{Et}):**

$$\begin{aligned} L_{Et} &= \frac{1}{N} \sum L_{it} = \phi w_t + \eta_t \underbrace{\frac{1}{N} \sum \lambda_i}_{\lambda_E} + u_{Et} \\ &= \phi w_t + \lambda_E \eta_t + u_{Et} \end{aligned}$$

3. **The Resulting GIV:** Back

$$z_t^{simple} = L_{St} - L_{Et} = (\lambda_{\mathbf{S}} - \lambda_{\mathbf{E}})\eta_{\mathbf{t}} + (u_{St} - u_{Et})$$

Econometrics Refresher

OLS and the Annihilator Matrix

Standard OLS projections decompose a vector y into explained and unexplained parts:

- **Projection Matrix (P):** $P = X(X'X)^{-1}X'$. Projects y onto the column space of X. $\hat{y} = Py$.
- **Annihilator Matrix (M):** $M = I - P$. Projects y onto the space orthogonal to X.
- **Orthogonality:** $MX = 0$. This ensures the residuals $e = My$ are uncorrelated with the regressors X.
- **Residuals:** $e = y - \hat{y} = (I - P)y = My$.

Notation Map:
Slide $\lambda \equiv$ Standard X (Regressors/Factors)
Slide $Q \equiv$ Standard M (Residual Maker Matrix)
Slide $S \equiv$ Standard y (Data Vector)
Slide $\Gamma \equiv$ Standard e (Residuals/Granular Weights) Back

Appendix B

Propositions

Orthogonal weights isolate idiosyncratic shocks

The GIV z_t is constructed from observables, L_{it}, using weights $\Gamma \in \mathbb{R}^N$ orthogonal to factor loadings:

$$z_t := \Gamma' L_t = \sum_{i=1}^{N} \Gamma_i L_{it}$$

By construction, this isolates a linear combination of idiosyncratic shocks u_t:

$$z_t = \Gamma' u_t \quad \Rightarrow \quad \text{exogenous to aggregate shocks } \eta_t, \varepsilon_t$$

This allows us to construct the sample estimators:

$$\hat{\psi}_T = \frac{\sum_{t=1}^{T} w_t z_t}{\sum_{t=1}^{T} L_{St} z_t} \qquad \hat{\phi}_T^d = \frac{\sum_{t=1}^{T} L_{\tilde{E}t} z_t}{\sum_{t=1}^{T} w_t z_t}$$

Back

Moments identify elasticities

The estimators follow directly from the exogeneity of z_t to structural shocks:

Inverse Supply Elasticity

From $w_t = \psi L_{St} + \cdots + \varepsilon_t$:

$$\mathbb{E}[(w_t - \psi L_{St}) z_t] = 0$$

$$\Rightarrow \quad \psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]}$$

Demand Elasticity

For weights $\tilde{E}$ where $\mathbb{E}[u_{\tilde{E}t} u_{\Gamma t}] = 0$:

$$\mathbb{E}[(L_{\tilde{E}t} - \phi^d w_t) z_t] = 0$$

$$\Rightarrow \quad \phi^d = \frac{\mathbb{E}[L_{\tilde{E}t} z_t]}{\mathbb{E}[w_t z_t]}$$

Back

Proposition 1

Under standard regularity conditions, as the number of time periods $T \to \infty$:

Inverse Supply Elasticity

$$\hat{\psi}_T \xrightarrow{\text{a.s.}} \psi$$

Demand Elasticity

$$\hat{\phi}^d_T \xrightarrow{\text{a.s.}} \phi^d \quad (\text{if } \psi \neq 0)$$

Intuition: The orthogonal weighting Γ ensures the instrument is uncorrelated with the structural error term, resolving simultaneity bias.

Back

Multipliers capture amplification

To understand the variance of these estimators, we define two key multipliers:

Quantity Pass-through

$$M := \frac{1}{1 - \psi\phi^d}$$

Price Pass-through

$$\mu := \psi M$$

These represent the pass-through from idiosyncratic and aggregate shocks to the aggregate quantity and the price, respectively.

Back

Proposition 2

The estimators converge in distribution to a normal distribution:

$$\sqrt{T}(\hat{\psi}_T - \psi) \xrightarrow{d} \mathcal{N}(0, \sigma_\psi^2), \quad \sigma_\psi = \frac{\mathbb{E}[u_{\Gamma t}^2]^{1/2}}{|\mathbb{E}[u_{St} u_{\Gamma t}]|} \frac{\sigma_{\varepsilon^\psi}}{|M|}$$

$$\sqrt{T}(\hat{\phi}_T^d - \phi^d) \xrightarrow{d} \mathcal{N}(0, \sigma_{\phi^d}^2), \quad \sigma_{\phi^d} = \frac{\mathbb{E}[u_{\Gamma t}^2]^{1/2}}{|\mathbb{E}[u_{St} u_{\Gamma t}]|} \frac{\sigma_{\varepsilon^\phi}}{|\mu|}$$

Back

Volatility improves precision

The variance equations reveal the signal-to-noise ratio inherent to GIV:

Signal Strength

$|\mathbb{E}[u_{St}u_{\Gamma t}]|$

Larger granular shocks increase relevance.

Instrument Noise

$\mathbb{E}[u_{\Gamma t}^2]^{1/2}$

Variance of the constructed GIV itself.

Structural Noise

$\sigma_\varepsilon/|M|$

Unobserved shocks dampening precision.

Back

Proposition 3

To minimize the asymptotic variances σ^2_{ψ} and $\sigma^2_{\phi^d}$, we use the $N \times N$ projection matrix Q that is orthogonal to the factor loadings λ:

$$Q := I - \lambda(\lambda'\lambda)^{-1}\lambda'$$

The optimal weights vector Γ^* is the projection of the size vector S:

$$\Gamma^* = S'Q$$

Result: Γ^* is the vector closest to S, while remaining orthogonal to the unobserved aggregate factor loadings.